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8

Experimental Methods and Real-World Behavior

8.1 INTRODUCTION

Up to this point we've been concerning ourselves with deterministic vibrations, i.e., periodically excited systems. The nice thing about periodic forcings is that once you know one period of the input, you know them all. Nothing else can happen—the input is completely determined. Thus we call the excitation deterministic. Although we could continue to restrict ourselves in this way, it turns out that expanding the scope of our excitations to include random vibrations will be very beneficial. For one thing, many systems are subjected to some form of random loading. The wind loads on an airliner, the forces acting on an offshore oil rig, and the accelerations felt by an object in a delivery truck are all well described by a random signal, not a periodically varying one. Furthermore, many of the algorithms used by modal analyzers to determine a system's natural frequencies and mode shapes are based on a random vibrations viewpoint. This chapter presents some of the important information you will need if you ever plan to carry out modal analyses, but please remember that it is just an introduction. The field of random vibrations is quite large, and there is a great deal of material that we won't be considering. Those interested in studying random vibration and modal analysis in more depth can consult other texts [3,9].

We'll start the chapter by laying out some fundamentals of random vibrations and finish by introducing some of the concepts of modal analysis.

8.2 SIGNAL DESCRIPTIONS

Let's begin this section by asking what a random signal is in the first place. Actually, you probably have a feel for what a random signal is, even if you haven't thought about it in a technical sense. Random signals look irregular, with no discernible pattern, like the example shown in Figure 8.1. At a simple level, we can think of random signals as waveforms that never repeat themselves. Thus a sinusoid isn't random because it repeats every T seconds, where T is the period of the oscillation. This is, however, not an adequate definition. Obviously, we wouldn't want to call $x(t) = at$ a random signal, even though it doesn't repeat itself. It's intuitively clear that $x(t) = at$ isn't random; it's just a constantly increasing signal. This is an example of a nonstationary situation. By "stationary," we mean that one or more averaged quantities remain fixed. For instance, in Figure 8.1 we see that the average value of x is zero; the signal seems to be centered about $x = 0$. Thus this signal exhibits stationarity. Signals that aren't stationary will drift, with time, as at does. Generally we will be concerned with stationary processes.

Even if the signal is stationary, nonperiodicity isn't enough. For example,

$$f(t) = \sin(t) + \sin(\sqrt{2}t)$$

isn't periodic but it certainly isn't random, as Figure 8.2 shows. Adding two sine waves whose frequencies aren't rationally related will always produce a nonperiodic result. We know that each individual sinusoidal signal isn't random, and thus adding two sinusoidal signals together shouldn't be considered to be random, since the function's value at a time t is completely predictable—we know it is simply composed of two sinusoids. This observation leads to the idea that random signals cannot

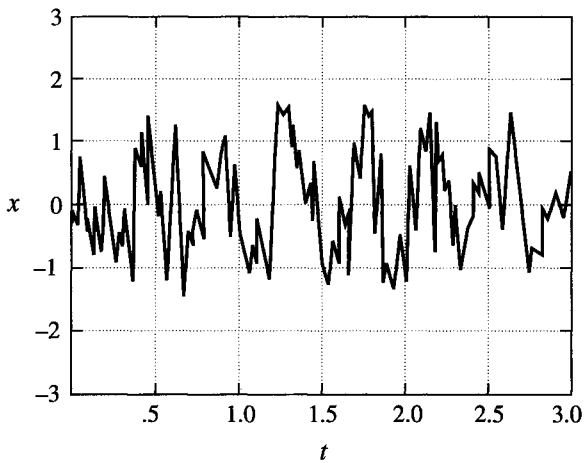


Figure 8.1 Random time response

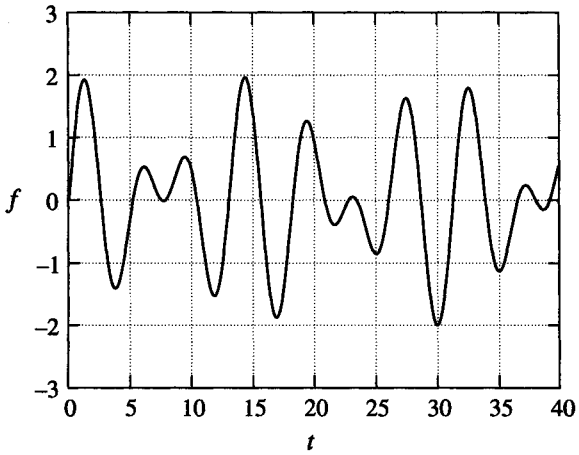


Figure 8.2 Response of two sinusoids with nonrationally related frequencies

be broken down into purely harmonic components. Thus we won't be able to find a Fourier series representation of a truly random signal.

Although people like to talk about random signals as if they truly exist, as far as our *computations* are concerned, they don't. Even if we had a steady state, truly random signal, one that continually and unpredictably changed over time, we'd have to cut off the data stream eventually. At this point our analyses can go one of two ways. We can view the actual signal as being zero from $t = -\infty$ to 0, equal to our data from $t = 0$ to $t = t_1$, and equal to zero for time greater than $t = t_1$, as shown in Figure 8.3. In this case the actual signal we're concerned with is a transient one, not a continuously varying, aperiodic one, and we obtain it by simply truncating the actual signal.

The other option is to impose a periodicity on the signal by assuming that the segment we captured repeats endlessly, with a period equal to t_1 (Figure 8.4). Just as in the preceding case,

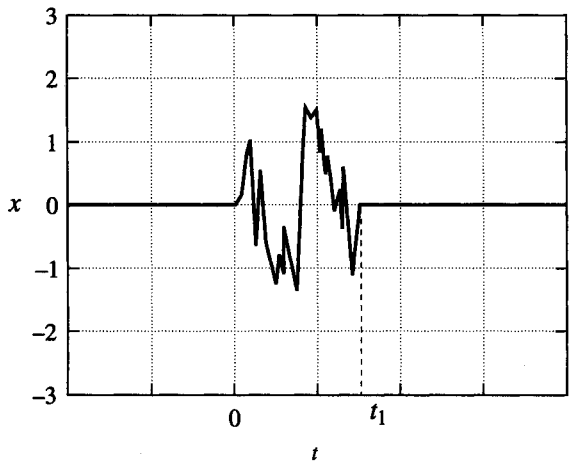


Figure 8.3 Truncated random time response

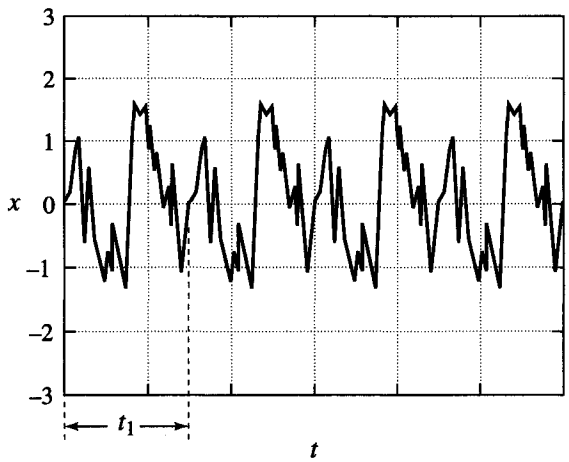


Figure 8.4 "Periodic" response

this viewpoint is not completely accurate. Whereas before we truncated the signal (making it into a transient), now we've made the signal periodic. Thus we can analyze either transient or periodic signals. The trick is to realize that as we increase the time t_1, we should be *approaching* the actual random signal. If we're lucky, these approximations will asymptotically approach the true random signal and we'll be left with only a small error for large enough t_1. It is important for the reader to keep these points in mind as we introduce the notions of random signals later in this chapter. Just as it was mathematically convenient to deal with viscous damping rather than the more realistic hysteretic or dry friction damping models, so it is more convenient to assume truly random signals, even when we know our analyses will actually be based on nonrandom ones.

You may have noted the use of the terms "random process" and "random signal" in the foregoing paragraphs. It is now time to ask what these words mean. Technically, there can be only random processes. It really makes no sense to call an acquired signal random since, once it has been acquired, it is completely determined. There isn't any randomness at all because the value of the signal for each point in time has been recorded, and there is no surprise about what value might occur for a given moment of time. Having said this, it's clear that this is a pretty pedantic viewpoint. Given a signal such as that shown in Figure 8.1, if you covered over the graph for t greater than 1 second, you would be hard pressed to reproduce the graph from a knowledge of the signal from zero to 1 second. That's what we really mean by random: that you can't predict what the signal will be at a later time, even if you know exactly what it was in the past. This is quite different from the case of a sinusoid, for which we know what all future values will be once we know the period and amplitude of the signal.

When we say random, what we're really concerned with are random *processes*, processes that generate an output that's unpredictable. Luckily, even if a random process produces an output that isn't predictable in the fine details, the outputs are nevertheless quite predictable in an averaged sense. For instance, if a skylight was opened during a rainstorm, you could pretty well predict that the floor beneath the skylight will be wet while the floor away from it will have stayed dry. Furthermore, you would expect the amount of water to be uniform, on average, over the wetted surface (this is ignoring

drips from the sides of the skylight, of course). You won't be able to predict exactly where the next drop of water will fall, but you'll be able to predict the drop's general location, and you'll also be able to predict the *average* amount of rain per unit time, given your observations of how much has been falling over the preceding minutes.

If the average amount of water per unit time stays constant, then the process is stationary. If the average amount of water per unit time increases (the rainfall moves from a gentle sprinkle to a thunderstorm), then the process is nonstationary. Obviously the property of stationarity depends upon our viewpoint. The rain will eventually stop; therefore it can't be strictly stationary. Thus stationarity depends on the time span of interest. If the average rainfall stays constant over a 5-minute span and we're not interested in times greater than this, then for our purposes we can consider the process to be stationary. This kind of view, looking at averages rather than fine details, is the key to describing random processes.

The most common signal descriptor we'll deal with is the mean value of a signal, given by

$$E(y) = \frac{1}{n}\sum_{i=1}^{n} y_i \tag{8.2.1}$$

Note that $E(y)$ is used to indicate the *expected value* (also called the *mean value*) of a variable. If we were concerned with a continuous process $y(t)$, we would have defined the mean as

$$E(y(t)) = \frac{1}{t_1}\int_0^{t_1} y(t)dt \tag{8.2.2}$$

where our data sample has a duration of t_1 seconds. The only problem with such a representation is that we will almost never be dealing with continuous signals. All real signals must be sampled, and the samples must be processed. Granted, a continuous data stream is more familiar and is sometimes easier to work with mathematically (just as viscous damping is easier than hysteretic), but the sad fact is that all our processing will necessarily be digital. Thus we'll often be concerned with a digital outlook.

Certainly the mean of a signal gives us important information. If we know the dynamic pressure loading on an airplane wing, then the mean of this loading gives us the static pressure the wing must be designed to withstand. The mean, however, isn't usually enough. Consider Figure 8.5, in which two different force loadings are shown, each on having a mean value of 10. Clearly these loadings are very different, even though that difference doesn't show up in the mean value. The signal in Figure 8.5*a* has a much lower amplitude as well as a lower frequency than that of Figure 8.5*b*. Thus, although the mean value is important, we can see that more information is needed.

The next descriptor we need to construct is one that will capture how much the signal deviates from its mean value. One choice would be to calculate

$$\sum_{i=1}^{n}[y_i - E(y)] \tag{8.2.3}$$

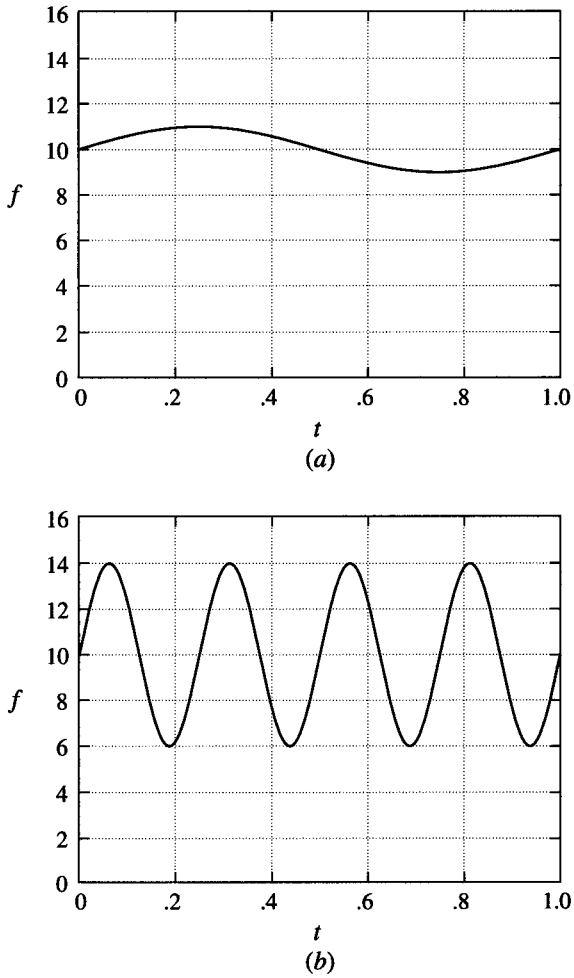


Figure 8.5 Same averaged behavior, but different dynamic behavior

Although this perhaps the most obvious choice, it's actually quite a poor one. For instance, this calculation would yield zero for both signals shown in Figure 8.5. This is because we've allowed both positive (when $y_i > E(y)$) and negative (when $y_i < E(y)$) values, which can act to cancel each other out. What we actually care about is how far the signal deviates from the mean value, and we aren't concerned about whether the deviation is positive or negative. So it is sensible to ensure that *all* deviations give a positive contribution to our descriptor. Although we could come up with a variety of different descriptors having this property, the one we'll introduce is the one used in common practice, called the *variance*:

$$\mathrm{var}(y) = \frac{1}{n}\sum_{i=1}^{n}[y_i - E(y)]^2 \tag{8.2.4}$$

You'll note that this could also be written as

$$\text{var}(y) = E[y - E(y)] \tag{8.2.5}$$

The variance is also called the *mean-square value.*

Often we'll talk about σ, the *standard deviation* of the signal

$$\sigma = \sqrt{\text{var}(y)} \tag{8.2.6}$$

which gives us a sense of exactly how far our signal is likely to be from the mean value (in an averaged way). This is also called the *root-mean-square* (RMS) *value* of the signal. If our signals are continuous in time rather than discrete, then the variance is found from

$$\text{var}(y) = \frac{1}{t_1}\int_0^{t_1} [y - E(y)]^2 \tag{8.2.7}$$

Although t_1 is always finite in any real analysis, we'll often set it to infinity for the purposes of our mathematical analyses.

Example 8.1

Problem Consider two different discrete signals. The first (s_a) is given by

$$s_a = 1, 1, 1, 1, 1, 13$$

and the second (s_b) is given by

$$s_b = 1, 5, 1, 5, 1, 5$$

Determine the expected value and variance of these signals.

Solution The expected value for both these discrete signals is 3. However the variances are quite different. For s_a we have

$$\text{var}(s_a) = \tfrac{1}{6}\left(2^2 + 2^2 + 2^2 + 2^2 + 2^2 + 10^2\right) = 20$$

while the variance of s_b is

$$\text{var}(s_b) = \tfrac{1}{6}\left(2^2 + 2^2 + 2^2 + 2^2 + 2^2 + 2^2\right) = 4$$

The variance for s_b is quite a bit smaller than that of s_a. This is because s_b varies only moderately from the mean value for all the entries, while s_a has a huge variation in its last entry. Thus we see that the variance allows us to distinguish between two signals as a function of how scattered the data are.

The foregoing descriptions have dealt with averaged quantities. It's possible to come up with many more averaged descriptors, but for our purposes mean and variance are sufficient. Next we'll want to look at a way of describing signals that depends on the particular value of the signal at particular times. Before doing this, however, we'll need to introduce the Fourier transform. The next section will describe this transform, and then we'll get back on track, discussing correlation functions and spectral density functions, both of which are important components of *spectral analyses.*

8.3 FOURIER TRANSFORM ANALYSIS

The basic building block of spectral analyses is Fourier transform analysis. As you'll recall from Section 3.2, we can take any signal of a fixed duration and determine the frequency components within it by calculating the Fourier series corresponding to the signal. The definition of a Fourier series was given by

$$x(t) = \sum_{n=0}^{\infty} a_n \cos(n\omega_0 t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_0 t) \tag{8.3.1}$$

for which

$$a_0 = \frac{\omega_0}{2\pi} \int_0^{\frac{2\pi}{\omega_0}} x(t)dt \tag{8.3.2}$$

$$a_n = \frac{\omega_0}{\pi} \int_0^{\frac{2\pi}{\omega_0}} x(t)\cos(n\omega_0 t)dt \tag{8.3.3}$$

and

$$b_n = \frac{\omega_0}{\pi} \int_0^{\frac{2\pi}{\omega_0}} x(t)\sin(n\omega_0 t)dt \tag{8.3.4}$$

Although this is approach is fine for periodic signals, random signals are decidedly *not* periodic. Indeed, we've already given, a working definition of a random signal as a signal that is not periodic and yet doesn't die down to zero. (Obviously, a signal that goes to zero isn't periodic but isn't necessarily random.)

If the signal isn't periodic, then we can't identify a fundamental frequency ω_0, and we can't use (8.3.1)–(8.3.4). So what do we do? One approach is to view the signal as we did in Figure 8.4. By artificially making the signal look periodic, we can apply the Fourier series approach and find its spectral components. It's clear that this approach will give more accurate approximations as the period we're using increases. To see where this approach ultimately leads us, we'll mathematically let the period go to infinity. This means that ω_0 goes to zero and the difference between the various harmonics also goes to zero. As this happens, our summation in (8.3.1) mutates into an integration. This isn't surprising when you realize that to simulate an integration on a computer we represent it as a finite summation. For more accuracy, we use more terms in the summation. Conceptually, we're running this in reverse here, going from a finite summation to an integration. The difference between the harmonics (ω_0) will become $d\omega$ (the differential frequency) and our individual harmonics won't be indicated by $n\omega_0$ anymore, but by ω. If you go through the details of this process you'll obtain

$$x(t) = \int_{-\infty}^{\infty} a(\omega)\cos(\omega t)d\omega + \int_{-\infty}^{\infty} b(\omega)\sin(\omega t)d\omega \tag{8.3.5}$$

for which

$$a(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(t)\cos(\omega t)dt \tag{8.3.6}$$

and

$$b(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(t)\sin(\omega t)dt \tag{8.3.7}$$

Equations (8.3.5)–(8.3.7) are one way to define the *Fourier transform* of a signal $x(t)$.

The other way of representing a Fourier transform is to use complex notation, just as we did when dealing with deterministic forced vibrations. If we let $X(\omega)$ represent the Fourier transform of $x(t)$, we'll have

$$x(t) = \int_{-\infty}^{\infty} X(\omega)e^{i\omega t}d\omega \tag{8.3.8}$$

and

$$X(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt \tag{8.3.9}$$

Complex and real forms are equally valid. For the complex case, $X(\omega)$ is complex and $x(t)$ is real, while in the real form $x(t)$, $a(\omega)$, and $b(\omega)$ are real.

Example 8.2

Problem Determine the Fourier transform of the signal shown in Figure 8.6. Consider both real and complex forms of the transform equations.

Solution Starting with the real form, we have

$$a(\omega) = \frac{1}{2\pi}\int_{0}^{a} \cos(\omega t)dt = \frac{1}{2\pi\omega}\sin(\omega a)$$

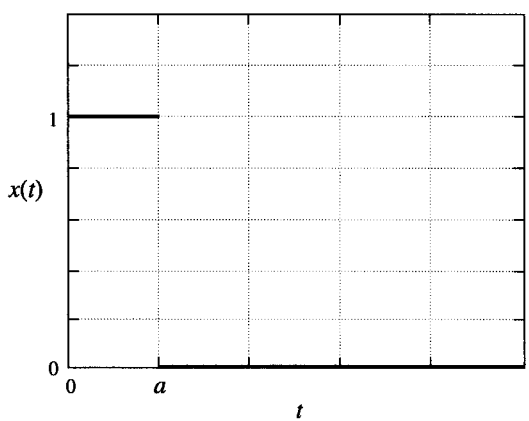


Figure 8.6 Rectangular pulse

and

$$b(\omega) = \frac{1}{2\pi}\int_0^a \sin(\omega t)dt = \frac{-1}{2\pi\omega}(\cos(\omega a) - 1)$$

Now let's consider the complex form. Equation (8.3.9) tells us that

$$X(\omega) = \frac{1}{2\pi}\int_0^a e^{-i\omega t}dt = \frac{-1}{2\pi i\omega}\left(e^{-i\omega a} - 1\right)$$

Expanding the complex exponential in terms of sine and cosine components gives us

$$X(\omega) = \frac{1}{2\pi\omega}(\sin(\omega a) + i(\cos(\omega a) - 1))$$

From this it seems like the real part of $X(\omega)$ is simply equal to the $a(\omega)$ we calculated from (8.3.6) and the imaginary part is equal to negative $b(\omega)$, found from (8.3.7).

To see that this wasn't just a coincidence, we can rewrite (8.3.9) as

$$X(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(t)\,(\cos(\omega t) - i\sin(\omega t))\,dt \tag{8.3.10}$$

Comparing (8.3.10) with (8.3.6) and (8.3.7), we see that the real part of the integral gives us $a(\omega)$ while the imaginary part yields $-b(\omega)$, thus supporting the validity of our calculated results

When Fourier transforms are first encountered they often seem advanced and difficult to understand. We've now seen, however, that they're easily viewed as simply limiting cases of a Fourier series, something you're probably quite comfortable with. In addition, you'll rarely have to worry about actually computing the Fourier transform in any real-world application; such calculations are carried out as internal functions in the *spectrum analyzers* you'll be using.

8.4 SPECTRAL ANALYSES

Now that we've seen what Fourier transform analysis is all about, we can get back to finding useful descriptors of our signals. The next approach we'll examine is not one that looks at an overall average; rather, it asks if there's any correlation between the signal at two distinct points in time. Clearly, a signal is always perfectly correlated with itself. But what if we ask whether there's a correlation between a signal at time t and at $t + \tau$? If the signal is a sine wave with period τ, then we know there should be an excellent correlation, since the responses at these two times will be identical. A purely random signal should be uncorrelated for any τ's besides zero, since the response changes in a completely unpredictable way. The autocorrelation function, $R_{xx}(\tau)$, deals with just this sort of question and is defined by

$$R_{xx}(\tau) \equiv \lim_{T\to\infty}\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)x(t+\tau)dt \tag{8.4.1}$$

We've already mentioned that infinitely long time traces aren't really practical; thus the limit in (8.4.1) really applies to T getting large, not truly infinite. Also, if our signal $x(t)$ has been truncated, the integral will have finite limits defined by where $x(t)$ goes to zero.

If $\tau = 0$, then (8.4.1) becomes

$$R_{xx}(0) \equiv \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t)dt \tag{8.4.2}$$

This is simply the mean-square value of $x(t)$. Thus the autocorrelation of a signal is equal to the signal's mean-square value when the delay $\tau = 0$.

A similar correlation function can be found by asking what the correlation is between two *different* signals, again with varying time delay τ:

$$R_{xy}(\tau) \equiv \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t)y(t+\tau)dt \tag{8.4.3}$$

This form, called the *cross-correlation*, is more general because we can always let y equal x if we want the autocorrelation.

Cross-correlations would be useful if you were trying to determine whether a vibration at one point of a structure was being influenced by vibrations at some other point. If the cross-correlation is high for some value of τ, you can deduce that there is a relation between the two vibrations and that the transit time for the motions to be transmitted from one point to the other is equal to τ. Cross-correlations are also used in radar applications. By sending out a signal and looking for the same signal delayed in time (because of being bounced off a plane that's being tracked), a radar unit can use the detected time delay and knowledge of the signal's transmission speed to determine how far away an aircraft is.

Most of the usefulness of these correlation functions for our purposes doesn't actually come directly from them but rather from functions that are defined from them. The first of these functions is known as the *power spectral density*, $S_{xx}(\omega)$, and is found from

$$S_{xx}(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} R_{xx}(\tau)e^{-i\omega\tau}d\tau \tag{8.4.4}$$

If you look back at (8.3.9) you'll note that the form shown in (8.4.4) suggests that the power spectral density is the Fourier transform of the autocorrelation function. Logically, we can therefore deduce that R_{xx} can be written as the inverse transform of $S_{xx}(\omega)$:

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} S_{xx}(\omega)e^{i\omega\tau}d\omega \tag{8.4.5}$$

The reason this is highly relevant is that the power spectral density does what its name implies. It tells us the density level of power that's present in a signal for any frequency of interest. If we want to know the total power between any two frequencies, then we need to integrate $S_{xx}(\omega)$ between these frequencies, taking into account both positive and negative ω's.

This particular descriptor is one of the cornerstones of spectral analysis. The outlook here is purely frequency based. Rather than worrying about the specific details of a signal in *time*, we're concerned with what's happening with respect to *frequency*. This can be made clear by an example.

Example 8.3

Problem To see how the power spectral density gives us a frequency-based insight into our analyses, determine $S_{xx}(\omega)$ by means of $R_{xx}(t)$ for

$$x(t) = a\sin(\omega_0 t)$$

Solution The autocorrelation is found from

$$R_{xx}(\tau) \equiv \lim_{T\to\infty} \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} a\sin(\omega_0 t) a\sin\left(\omega_0(t+\tau)\right) dt$$

Expanding out the second sine term yields

$$R_{xx}(\tau) = \lim_{T\to\infty} \frac{a^2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \sin(\omega_0 t)\sin(\omega_0 t)\cos(\omega_0\tau)dt$$

$$+ \lim_{T\to\infty} \frac{a^2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \sin(\omega_0 t)\cos(\omega_0 t)\sin(\omega_0\tau)dt \tag{8.4.6}$$

The second integral goes to zero as T goes to infinity, while the first will leave us with

$$R_{xx}(\tau) = \frac{a^2\cos(\omega_0\tau)}{2} \tag{8.4.7}$$

This result makes sense because it tells us that the sine is maximally correlated with itself for delays of $\frac{2\pi}{\omega_0}$ (the times when $\cos(\omega_0\tau)$ equals 1.0) and is not correlated at all for a time shift of $\frac{\pi}{2\omega_0}$. Since a time shift of $\frac{\pi}{2\omega_0}$ alters a sine into a cosine (which we know is an independent function), we'd expect zero correlation.

Equation (8.4.7) predicts that the mean-square value $R_{xx}(0)$ is equal to $\frac{a^2}{2}$. We can verify that this is so analytically. If $x(t) = a\sin(\omega_0 t)$, then $x^2 = a^2\sin^2(\omega_0 t) = a^2\left(\frac{1}{2} - \frac{\cos(2\omega_0 t)}{2}\right)$. Thus, since the $\cos(2\omega_0 t)$ will average out to zero, we're left with the mean-square value being $\frac{a^2}{2}$, just as predicted.

Now that we've got the autocorrelation function, we can compute $S_{xx}(\omega)$:

$$S_{xx}(\omega) = \lim_{n\to\infty} \frac{1}{2\pi}\int_{-\frac{n\pi}{\omega_0}}^{\frac{n\pi}{\omega_0}} \frac{a^2\cos(\omega_0\tau)}{2} e^{-i\omega\tau} d\tau$$

You'll notice that this is a bit different from (8.4.4) in that it involves a limit. If we immediately tried to use the infinite limits of integration, we'd be unable to evaluate the integral. By taking the time integral to be integer multiples of the autocorrelation's period (using $\frac{n\pi}{\omega_0}$ instead of ∞), and letting the number of multiples go to infinity, we'll get the correct final solution and also be able to see the way in which we approach this solution.

Carrying out the foregoing integrations will leave us with

$$S_{xx}(\omega) = \lim_{n\to\infty} \frac{1}{4\pi}\left(\frac{a^2}{\omega_0 - \omega}\sin\left(n\pi - \frac{n\pi\omega}{\omega_0}\right) + \frac{a^2}{\omega_0 + \omega}\sin\left(n\pi + \frac{n\pi\omega}{\omega_0}\right)\right) \tag{8.4.8}$$

Figure 8.7 shows what this function looks like for different values of n. Only ω values greater than zero are shown. Since $S_{xx}(\omega)$ is an even function (which we haven't proved but is true nonetheless), there's no need to plot it for negative ω values—the behavior is identical to that for positive ω's. For $n = 4$ (Figure 8.7*a*) we see that a spike is developing at $\omega = 5$ rad/s (for this plot ω_0 was chosen to be 5). If we increase the integration duration by a factor of 20 (bringing n to 80) we get Figure 8.7*b*. What you can see from these two plots is that as n is increasing, the spike at $\omega = 5$ is also increasing. We can determine this analytically in a straightforward manner. Let $\omega = \omega_0 - \varepsilon$ in (8.4.8). The second term isn't going to be important because the first is going to be completely dominate the response. The first term becomes

$$S_{xx}(\omega) = \frac{a^2}{2\pi}\lim_{n\to\infty}\frac{\sin\left(\frac{n\pi\varepsilon}{\omega_0}\right)}{\varepsilon}$$

As we look at smaller and smaller values of ε, we obtain an indeterminant fraction. Expanding the sine in terms of its trigonometric representation (or using L'Hôspital's rule) gives us

$$S_{xx}(\omega_0) = \lim_{n\to\infty}\frac{na^2}{2\omega_0}$$

Clearly, as n goes to infinity, so does $S_{xx}(\omega)$.

Everything away from $\omega = \omega_0$ or $\omega = -\omega_0$ goes to zero, and we're left with two Dirac delta functions, one located at $\omega = \omega_0$ and the other at $\omega = -\omega_0$. The value of each Dirac delta function turns out to be .25, i.e., in the limit of $n \to \infty$ we have Figure 8.7*c*

$$S_{xx}(\omega) = \frac{a^2}{4}\delta(\omega + \omega_0) + \frac{a^2}{4}\delta(\omega - \omega_0) \tag{8.4.9}$$

This is a very relevant finding. It tells us that all the power in the sine wave is concentrated at one frequency. Thus when we run modal tests and find sharp power spectral density peaks, we'll know it means that there are distinct harmonics in the signal under analysis and that the frequencies of the harmonics correspond to the frequencies of the peaks. Note that we got this frequency-based information from an analysis of a time-based function (the autocorrelation function). The whole field of spectral analysis revolves around this viewpoint that's based in the frequency domain rather than in time.

A tremendously useful result, which we won't derive but shall simply present, is the following:

$$S_{xx}(\omega) = |g(\omega)|^2 S_{ff}(\omega) \tag{8.4.10}$$

where $g(\omega)$ is the transfer function of the system under consideration, $f(t)$ is the input to the system, and $x(t)$ is the system's output. $S_{xx}(\omega)$ is the power spectral density of the output, and $S_{ff}(\omega)$ is the power spectral density of the input.

For example, assume that our equation of motion was

$$m\ddot{x} + kx = f(t) \tag{8.4.11}$$

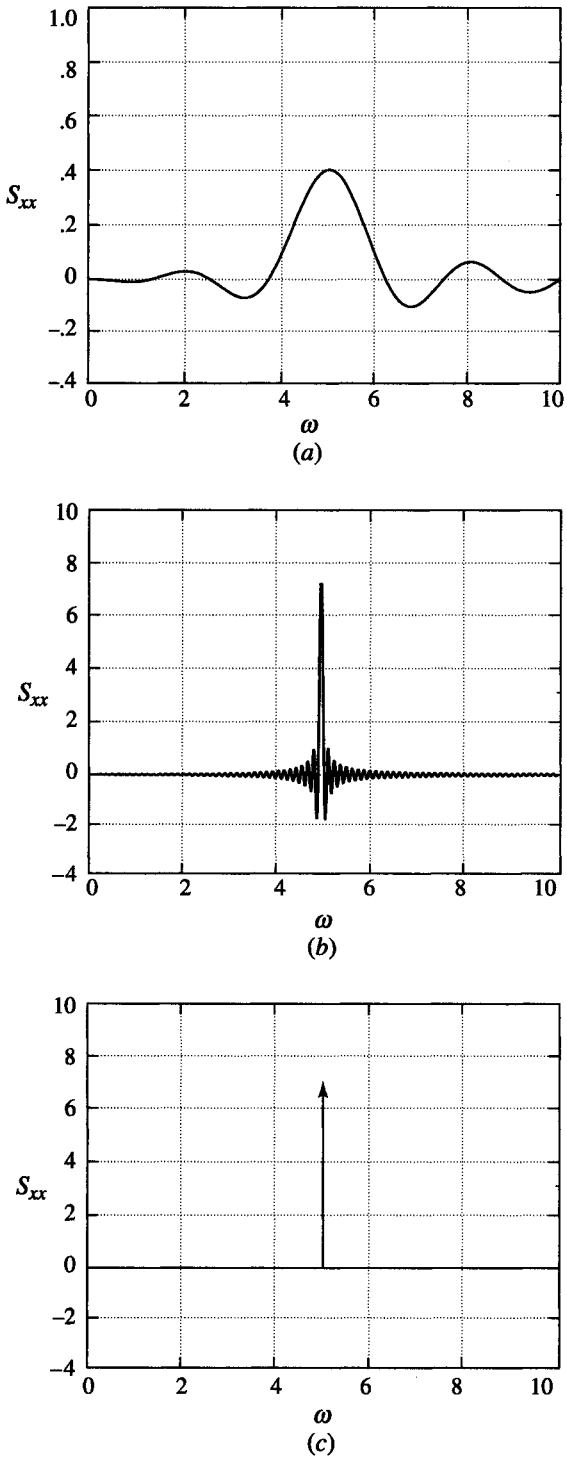


Figure 8.7 Power spectral density of a 5 rad/s sine wave with increasing numbers of cycles included in the calculation

in which case the transfer function is

$$g(t) = \frac{1}{k - m\omega^2} \tag{8.4.12}$$

If we know what the power spectral density of the forcing is, then we can immediately calculate the power spectral density of the output from (8.4.10) as

$$S_{xx}(\omega) = \left| \frac{1}{k - m\omega^2} \right|^2 S_{ff}(\omega) \tag{8.4.13}$$

Equation (8.4.10) can be used in a number of ways. The most obvious was just discussed, namely, finding the response power spectral density from knowing the system transfer function and the forcing power spectral density. But think about how else we can use it. One other use would be to determine the input excitation to a system by measuring the output power spectral density and dividing it by the square of the transfer function. This might be useful if we can't easily determine the input directly but have access to the system output and know the system characteristics.

This equation is most commonly used as a means of actually determining the system characteristics simply by measuring the input and output signals. Dividing (8.4.10) by $S_{ff}(\omega)$ gives us

$$|g(\omega)|^2 = \frac{S_{xx}(\omega)}{S_{ff}(\omega)} \tag{8.4.14}$$

Thus

$$|g(\omega)| = \sqrt{\frac{S_{xx}(\omega)}{S_{ff}(\omega)}} \tag{8.4.15}$$

Obtaining the power spectra of the input and output signals is usually easy. Common transducers for acquiring data of this sort would be an impact hammer to excite the system and produce an input signal and an accelerometer to produce output data. Both signals would be fed into a modal analyzer, which would calculate the system's transfer function in a manner similar to that just shown. Of course, in a real experimental setup there will be noise in the measurements, and the analyzer won't carry through exactly as shown in (8.4.15). The general approach, however, is valid and the differences between the analyzer's algorithms and (8.4.15) would simply be modifications introduced to minimize computational errors due to experimental error sources. Further information on this subject can be found elsewhere [3] and also (briefly) later in this section. One interesting point is that (8.4.15) doesn't include any phase information for the system's transfer function. This information can be found with a slightly different approach. Again, details are available [3].

Example 8.4

Problem As an example of how the foregoing approach is used, we'll look at a problem we already know how to solve, namely, the response of a spring-mass under sinusoidal forcing. Consider the system shown earlier in Figure 2.10. Let the input force be $2\sin(\omega t)$ N, $m = 2$ kg, and $k = 100$ N/m. For this example we'll let $\omega = 10$ rad/s. Use two approaches to determine the mean-square value of the response, and compare the results. First, determine the power spectral density of the output, find the autocorrelation function from this, and evaluate the autocorrelation at $\tau = 0$ (giving the mean-square response). Next, solve the problem exactly and simply calculate the mean-square response from the known solution. Both methods should yield the same answer.

Solution The transfer function for this problem is given by

$$g(\omega) = \frac{1}{k - m\omega^2}$$

The power spectral density of the input is given by (8.4.9)

$$S_{ff}(\omega) = \frac{2^2}{4}\left(\delta(\omega + 10) + \delta(\omega - 10)\right)$$

Using (8.4.10) we find

$$S_{xx}(\omega) = \left|\frac{1}{k - m\omega^2}\right|^2 \left(\delta(\omega - 10) + \delta(\omega + 10)\right)$$

Substituting the values for m, k, and ω yields

$$S_{xx}(\omega) = \frac{1}{10{,}000}\left(\delta(\omega - 10) + \delta(\omega + 10)\right)$$

We can next take the inverse transform of this to find the autocorrelation $R_{xx}(\tau)$:

$$R_{xx}(\tau) = \frac{1}{10{,}000}\int_{-\infty}^{\infty}\left(\delta(\omega - 10) + \delta(\omega + 10)\right)e^{i\omega\tau}\,d\omega$$

which yields

$$R_{xx}(\tau) = \frac{1}{5000}\cos(10\tau)$$

Evaluating this at $\tau = 0$ gives us

$$R_{xx}(0) = \frac{1}{5000} \tag{8.4.16}$$

Now let's see if this matches the result we'd get from calculating the actual forced response and evaluating the mean-square value from it. The amplitude response of the system is given by

$$\bar{x} = \frac{\bar{f}}{k - m\omega^2}$$

where $\bar{x}$ is the response magnitude and $\bar{f}$ is the forcing magnitude. Substituting the values for this example yields

$$\bar{x} = \frac{2}{100 - 2(100)} = -\frac{1}{50}$$

Thus the total response is given by

$$x(t) = -\frac{1}{50}\sin(10t)$$

The mean-square value of this is given by

$$E(x^2) = \left(\frac{1}{50^2}\right)\left(\frac{1}{2}\right) = \frac{1}{5000}$$

Note that this agrees with (8.4.16). We got the same answer by directly determining the mean-square value from a knowledge of the actual response, and by deducing it from a knowledge of the response's power spectral density.

We've now seen that the transfer function of a system relates the input power spectral density and the output power spectral density,

$$S_{xx}(\omega) = |g(\omega)|^2 S_{ff}(\omega) \tag{8.4.17}$$

Two other formulas that are useful in modal analysis, which were alluded to a page or so ago, are

$$S_{fx}(\omega) = g(\omega)S_{ff}(\omega) \tag{8.4.18}$$

and

$$S_{xx}(\omega) = g(\omega)S_{xf}(\omega) \tag{8.4.19}$$

where the xf and fx indicate that these are power spectral densities found from the cross-correlation functions between the input and output. We can estimate the system transfer function in two more ways now. The first estimate, found from (8.4.18), is

$$g_1(\omega) = \frac{S_{fx}(\omega)}{S_{ff}(\omega)} \tag{8.4.20}$$

and the second, found from (8.4.19), is

$$g_2(\omega) = \frac{S_{xx}(\omega)}{S_{xf}(\omega)} \tag{8.4.21}$$

Noise in the measurements enters into these expressions in different ways, and they can be used together to find the best overall transfer function estimate. Normally, as a result of experimental problems, the estimates are not exactly the same. If we use the two estimates shown in (8.4.20) and (8.4.21), we can form their ratio:

$$\gamma^2 = \frac{g_1(\omega)}{g_2(\omega)} \tag{8.4.22}$$

where γ^2 is called the *coherence function*. This frequency-dependent function is a widely used measure of how "believable" our measurements are. We won't derive it here, but it can be shown that γ^2 always lies between 0 and 1. If the coherence is near zero, our measurements should be viewed with great suspicion, whereas a coherence near 1.0 indicates that the $g_1(\omega)$ and $g_2(\omega)$ are in very close agreement, which in turn implies that the measurements are believable and that noise and nonlinear effects aren't significant. Most analyzers will output the coherence whenever they compute a system transfer function, allowing the user to decide whether to accept or reject the computed results.

8.5 NOISE

Now that we've seen how to analyze general random inputs, it is time to examine the most common random signal around, namely *white noise*. Returning to Figure 8.1, you can see what a white noise signal might look like when plotted versus time—clearly, it is a very jagged function, not at all regular. It is so irregular, in fact, that it is completely unpredictable from one instant to the next. This means that its autocorrelation function is simpy a Dirac delta function, $\delta t - \tau$. For $\tau = 0$ the signal is completely correlated, and for any other τ it is completely uncorrelated. If we denote the magnitude of the Dirac delta function by w_0, then our signal's autocorrelation is given by

$$R_{xx}(\tau) = w_0\delta(\tau) \tag{8.5.1}$$

The power spectral density for this function is given by

$$S_{xx}(\omega) = \frac{w_0}{2\pi} \tag{8.5.2}$$

As you can see (Figure 8.8), the power spectral density is a constant, regardless of frequency. Thus the density of power is the same at all frequencies.

Does this make sense from a physical point of view? Well, if the output is a constant at all frequencies, then we can integrate over all frequencies to find the total power of the signal. Since

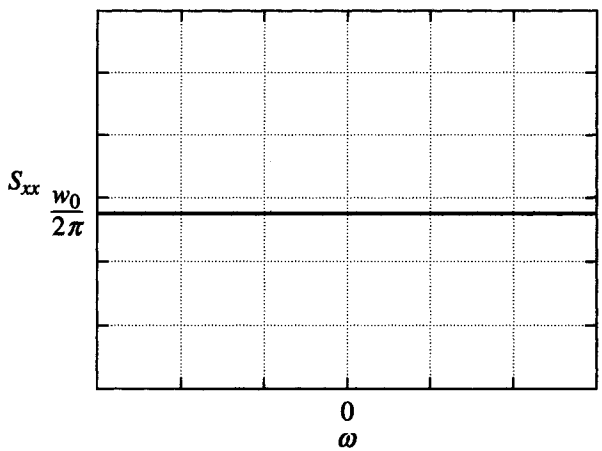


Figure 8.8 Constant power spectral density

the power spectral density has the same value everywhere and extends out to infinity, the integral is infinite. Thus the signal must have infinite energy. This is a problem. It's pretty obvious that we can't produce a signal that has an *infinite* amount of energy; all real signals must be finite.

What's the conclusion? Simply that our assumption of a constant power spectral density over all frequencies must be erroneous. At some point, the power spectral density has to drop off, as shown in Figure 8.9. This figure shows what all real-world, "white" signals really look like. They're reasonably constant over some range, and then the energy drops off to zero. An idealization of this situation is shown in Figure 8.10, in which we assume a constant level to ω_1 and then have an immediate cutoff to zero. Such signals are called *band-limited* noise.

Luckily, the difference between ideal white noise and actual white noise doesn't really affect our analyses. As we've already seen, the transfer function of our mechanical system will drop to zero as we pass its highest natural frequency. Thus, even if we had an ideal white noise input, the response

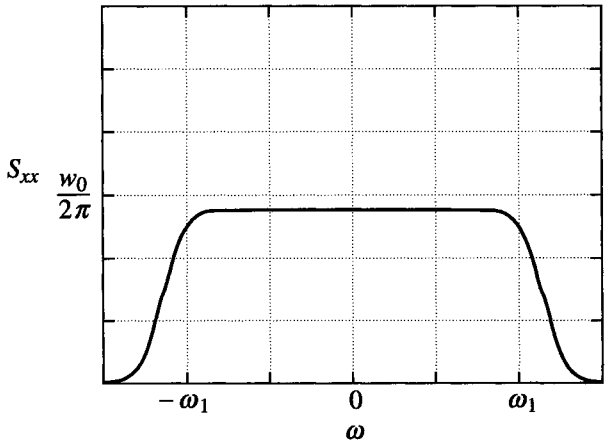


Figure 8.9 Power spectral density of band-limited white noise (gradual frequency cutoff)

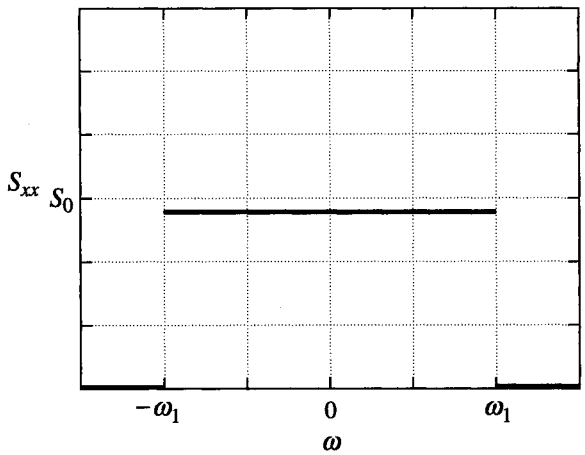


Figure 8.10 Power spectral density of band-limited white noise (sharp frequency cutoff)

will drop to zero at high frequencies because the transfer function itself is going to zero. We just have to make sure that the white noise signal is pretty much constant up to and a little beyond the highest natural frequency of the system of concern.

Another way of seeing how ideal white noise differs from what we find in real life is to realize that the autocorrelation won't really be a delta function. If the output in time was in fact changing in a completely random manner, by implication the acceleration experienced in the system would be infinite, since only an infinite acceleration could move the output from one state to another in an infinitesimal time span. Since we know that real systems are to some extent limited regarding the speed with which they can change, we know that an actual signal at $t = \tau$ will be close to its value at $t = 0$, for small τ. An "infinitely jagged" function (such as ideal white noise will have) requires energy at ultrahigh frequencies. This high frequency content is what lets the signal "wiggle" so quickly.

We can compute the autocorrelation corresponding to a more realistic, band-limited signal to see how cutting off the spectral energy at some finite frequency affects matters. Figure 8.10 shows our power spectral density. It is constant at a level S_0 from $-\omega_1$ to ω_1 and zero everywhere else. Calculating the autocorrelation for this case is straightforward. Recalling that for the general problem we have

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} S_{xx}(\omega) e^{i\omega\tau} d\omega \tag{8.5.3}$$

for our particular problem we write

$$R_{xx}(\tau) = S_0 \int_{-\omega_1}^{\omega_1} e^{i\omega\tau} d\omega = 2S_0 \frac{\sin(\omega_1 \tau)}{\tau} \tag{8.5.4}$$

This autocorrelation function is plotted in Figure 8.11. For this plot the value of S_0 was set equal to .5 and ω_1 was 50 rad/s. As you can see, the correlation drops off quickly, but not instantly. As τ increases, the autocorrelation goes to zero. Finally, the magnitude at $\tau = 0$ is finite, not infinite as would be the case for a Dirac delta function.

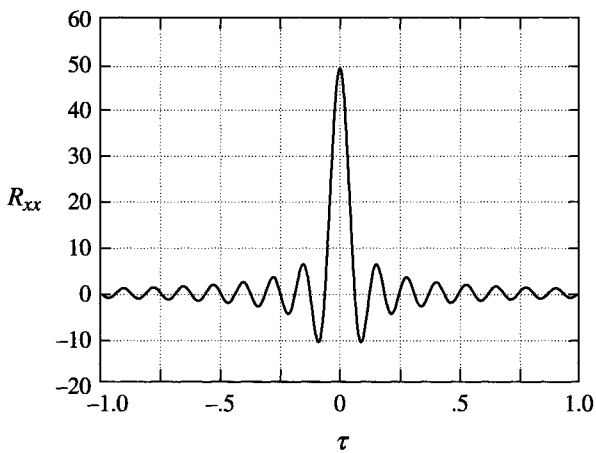


Figure 8.11 Autocorrelation of band-limited white noise

Figures 8.12–8.15 give an overview of the most basic responses we might expect to see. Figure 8.12 shows the relationship between white noise (a function of time), its autocorrelation (a function of time delay τ), and its power spectral density (a function of frequency). For white noise, the autocorrelation $R_xx(\tau)$ is a pure delta function and the power spectral density $S_{xx}(\omega)$ is a constant.

Real-world effects mean that the closest we'll ever get to Figure 8.12 is the situation shown in Figure 8.13. The time function looks similar to that of the preceding case. The only difference is that if we examined it in fine detail we'd see that the response of Figure 8.13*a* is continuous, while that of Figure 8.12*a* is discontinuous (implying finite changes of output for infinitesimal changes in time). The autocorrelation is therefore not a pure Dirac delta function, and the power spectral density drops off at high frequencies.

Figure 8.14 shows what we'll see if the signal is a pure sinusoid. The autocorrelation for this case is also sinusoidal (telling us that the signal is periodic), and the power spectral density is composed of Dirac deltas at the frequency of the input signal. As we've already found (Figure 8.7), this precise response won't ever be seen because we'd have to observe the input sinusoid for an infinitely long time to get a power spectral density that's in the form of a delta function. Furthermore, most real signals will have some noise superimposed on them, as shown in Figure 8.15*a*. The autocorrelation of this signal is shown in Figure 8.15*b*. The effect of the noise is to create a large response at $\tau = 0$, whereas the rest of the response is dominated by the sinusoidal signal (and thus gives us largely periodic behavior). Finally, Figure 8.15*c* shows the power spectral density. Since the noise is independent of the sinusoidal signal, we end up with the delta functions associated with sinusoids, along with a low level of constant energy density (associated with the noise).

8.6 SENSORS AND ACTUATORS

If you're going to run a modal test, a few basic ingredients must be present. You need some way of exciting the system, you need a way to sense the system's response, and you need a device that can use the input and output information to deduce some of the system properties. Let's start with the question of excitation. The two most widely used excitations are the *impact hammer* and the *shaker* (electrohydraulic or electromechanical).

The *impact hammer* (Figure 8.16) looks very much like an ordinary hammer and is used in a similar way. One holds the impulse hammer in much the same way as one holds a regular hammer and quickly strikes the structure being tested. Some care is necessary to make sure that the hammer hits the structure only once. (If the hammer isn't held correctly, the head may contact the structure two or more times in quick succession, ruining the data acquisition.) We saw in Section 3.4 that loadings that occur quickly look and act like delta functions. We've also found that delta functions have a constant power spectral energy. This characteristic is great for our analyses. Recall that the absolute value of the transfer function can be found from (8.4.15)

$$|g(\omega)| = \sqrt{\frac{S_{xx}(\omega)}{S_{ff}(\omega)}} \tag{8.6.1}$$

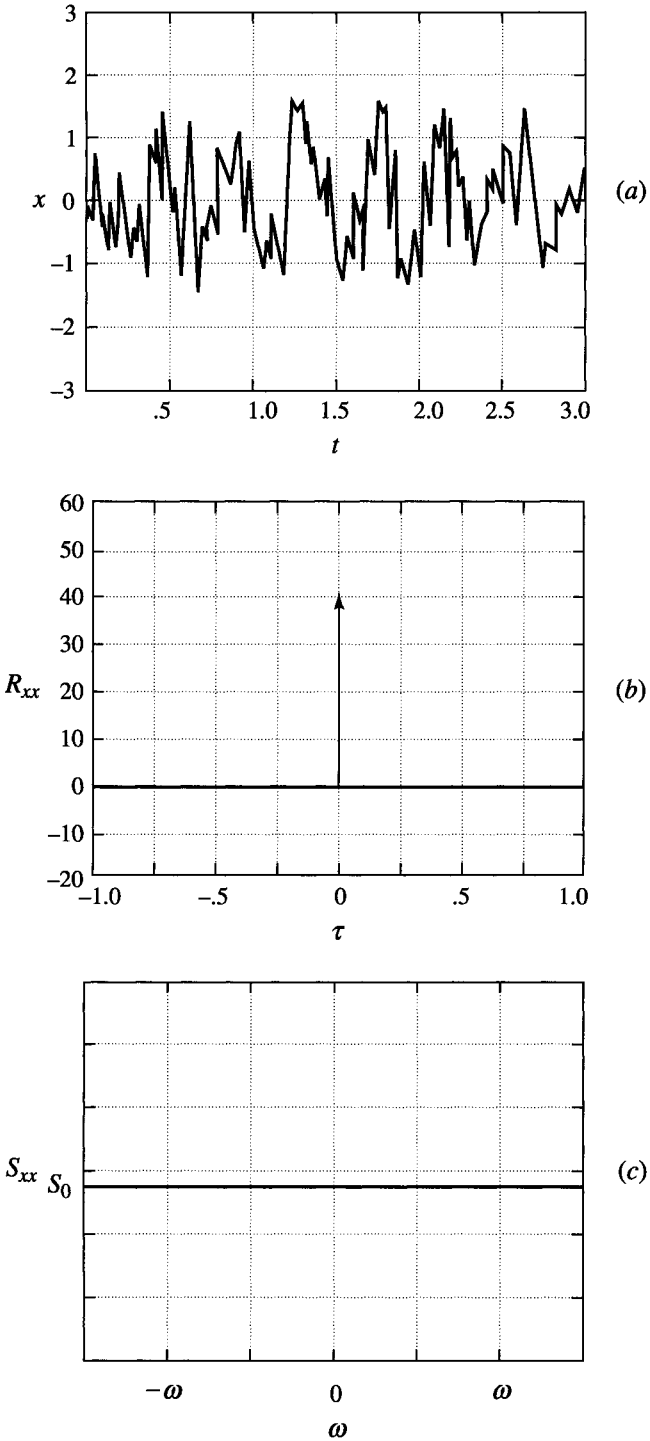


Figure 8.12 (*a*) Time trace, (*b*) autocorrelation, and (*c*) power spectral density for white noise

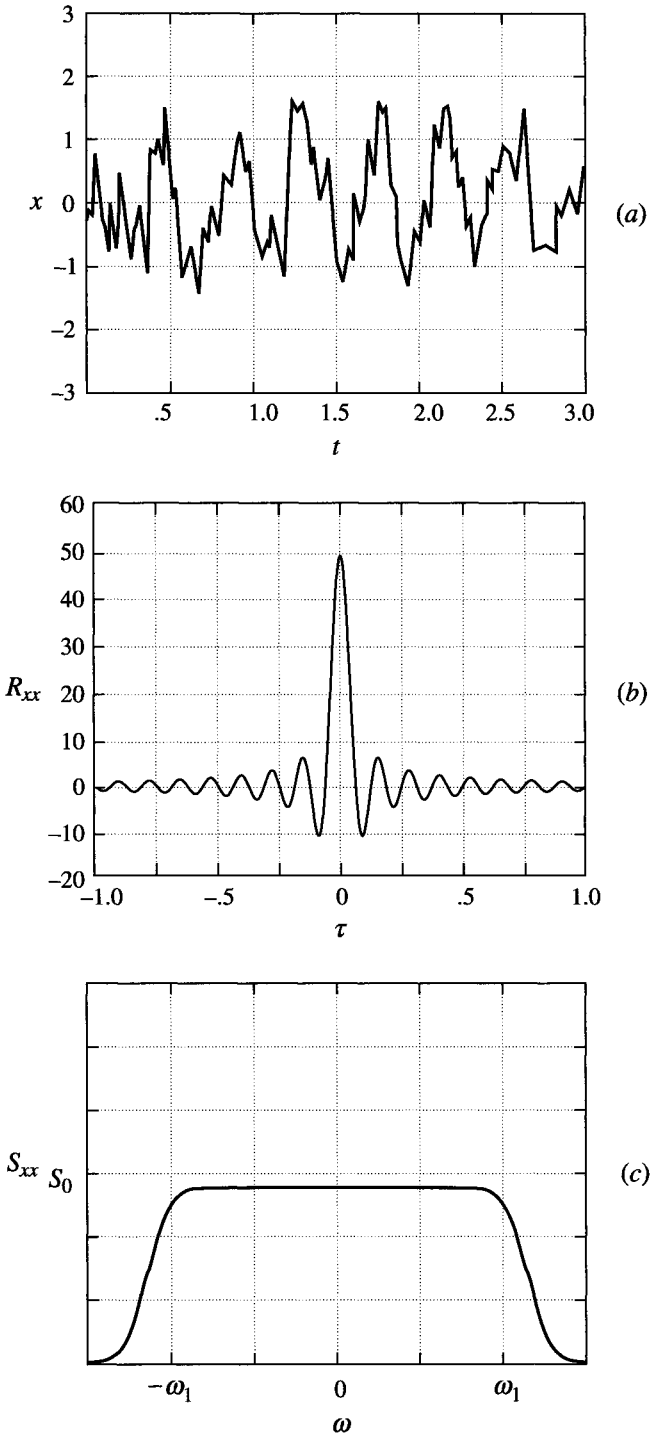


Figure 8.13 (*a*) Time trace, (*b*) autocorrelation, and (*c*) power spectral density for band-limited white noise

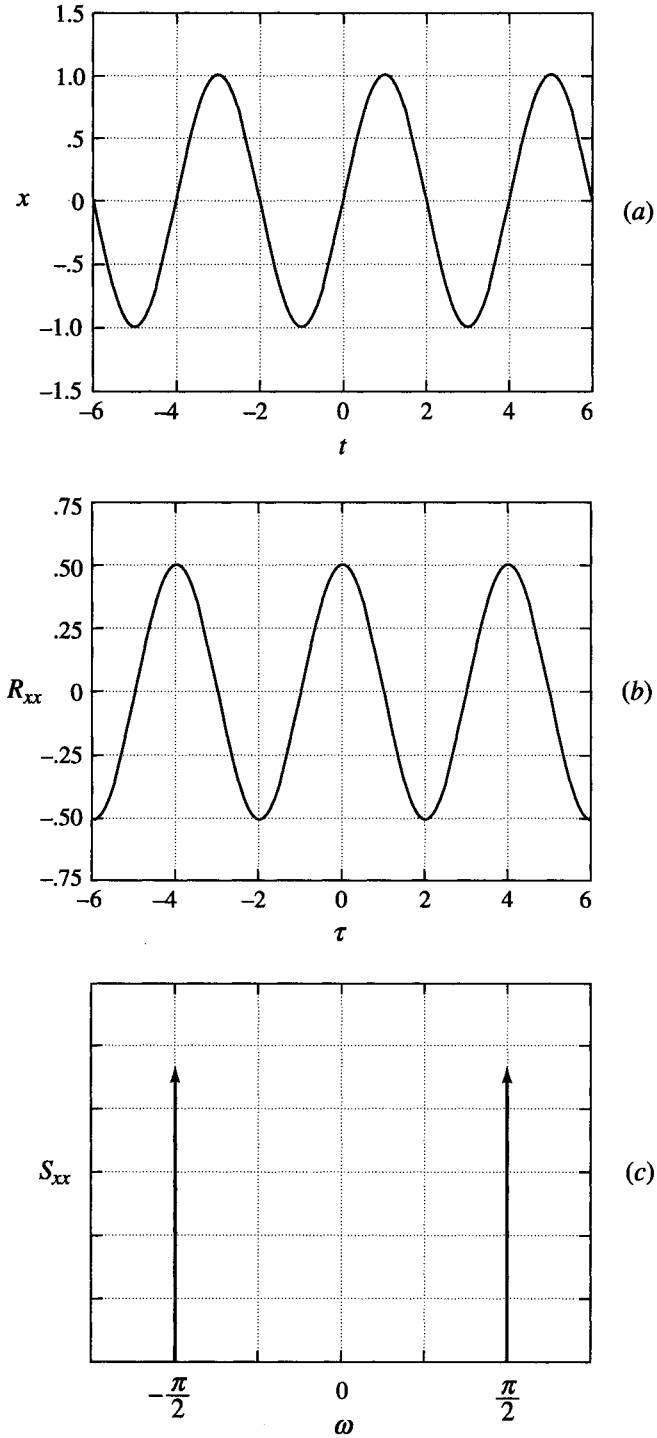


Figure 8.14 (*a*) Time trace, (*b*) autocorrelation, and (*c*) power spectral density for $\sin\left(\frac{\pi}{2}t\right)$

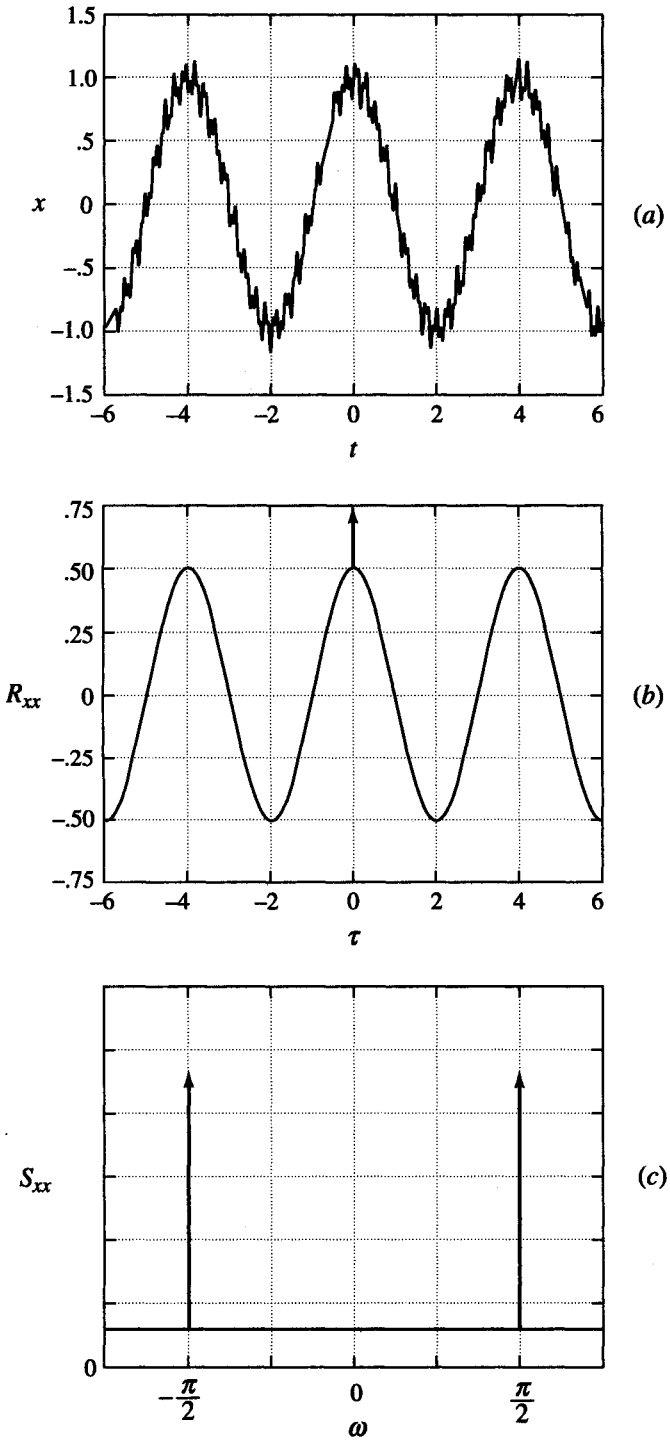


Figure 8.15 (*a*) Time trace, (*b*) autocorrelation, and (*c*) power spectral density for a combination of white noise and a sinusoidal signal

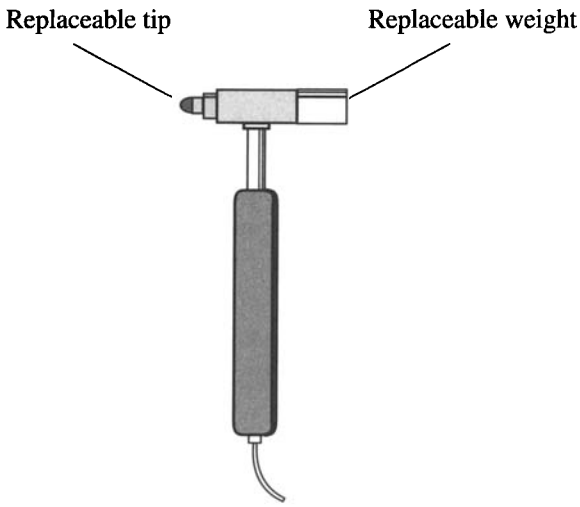


Figure 8.16 Impact hammer

If the input power spectral density is a constant (call it S_0), then we have

$$|g(\omega)| = \frac{\sqrt{S_{xx}(\omega)}}{\sqrt{S_0}} \tag{8.6.2}$$

Thus instead of having to take the ratio of two frequency-dependent functions, we need only divide the output's power spectral density by a constant. Actually, our analyzers will still calculate the ratio of the two power spectral densities, but since the input is essentially flat, calculation errors are minimized.

Just behind the tip of the head (the part striking the structure) is a piezoelectric sensor that generates a signal proportion to the impact force. By itself, this element is known as a *force transducer*. A cable runs from the hammer to a modal analyzer, supplying the output of the force transducer as an input to the analyzer. The analyzer can then determine what the precise force profile versus time was during the strike and from that calculate the power spectral density of the input force.

Impact hammers come with weights that can be added to the head, and different tips. The weights allow different force levels to be input to the structure, and the tip hardness determines how long the tip stays in contact with the structure (which affects the spectral energy input). Using a hard tip gives a wider frequency input but lowers the overall energy that's transferred to the structure. Thus the hard tips allow you to analyze higher frequency responses but might not supply enough excitation energy for accurate calculations. Of coure, softening the tip lets you put more energy in at the expense of a restricted excitation bandwidth.

The advantages of impact hammers are that they are fast to use and relatively inexpensive (on the order of $1000). In addition, they aren't attached to the structure and therefore don't alter its dynamic characteristics. They also have the advantage of being able to input energy at any point of

the structure you can get a clean hit on. One disadvantage is that you don't know *precisely* where you're hitting the structure, and thus you're bringing some uncertainty into your measurements. In addition, you might hit the structure too hard and damage it, but if you don't hit it hard enough, you may not be able to input enough energy to adequately excite the system.

The second type of excitation device is known as a *shaker*. These devices move the test structures mechanically (electromechanical shaker) or hydraulically (electrohydraulic shaker). The more common type is the electromechanical shaker (Figure 8.17). If you're familiar with a loudspeaker then you're also familiar with electromechanical shakers—there's really little fundamental difference between the two. The speaker cone moves because a voice coil is caused to vibrate owing to an alternating current. This is precisely what happens in an electromechanical shaker. The only difference is in the power and accuracy involved. Shakers must be capable of producing a significant time-varying force with which to excite the test structure (unlike loudspeakers, which merely have to move the speaker cone and a bit of air).

The shaker is physically connected to the test structure by means of a thin rod of plastic or metal, known as a stinger. The stinger is much stiffer along its length than laterally. Thus if it is placed in line with the shaker's axis of movement, it effectively transmits the shaker forces but doesn't cause unwanted twisting of the structure.

Because we are able to control the electrical signal reaching the shaker, we can produce any input waveform we like. The most obvious signal would be a sinusoidal one, allowing us to determine the structure's response at a single frequency. We could also input a random signal

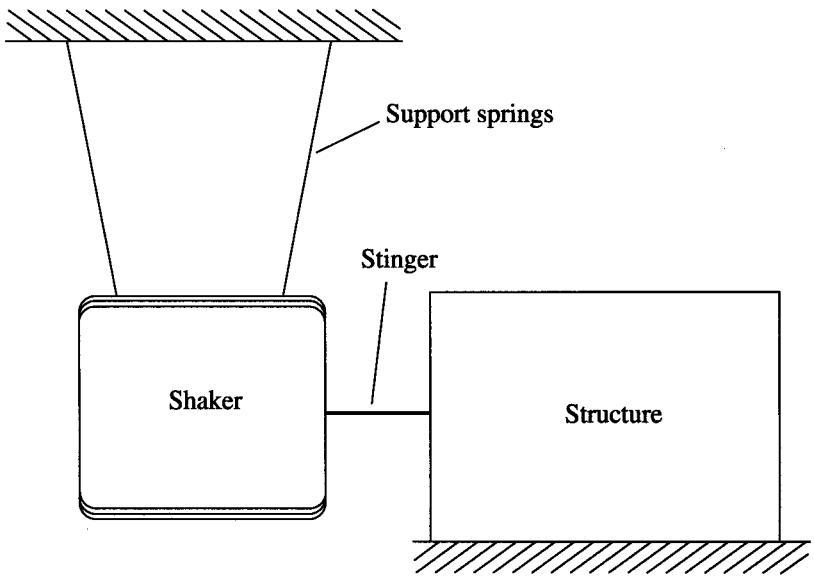


Figure 8.17 Elastically supported shaker

(band-limited white noise), which would let us calculate the entire frequency response of the system. Another common input is called a swept sine. This consists of a sinusoidal excitation that starts at low frequencies and ramps upward. This signal also allows us to determine the structure's complete frequency response. Transient inputs are also possible and are used to mimic shock loadings.

The advantages of a shaker are the variety of excitations that can be used, the range of force levels, and the accuracy with which the inputs can be applied. The disadvantages are cost (small shakers and associated power amps can run $2,000, with larger ones hitting $10,000–$20,000 without difficulty) and the fact that shakers modify the structures they are attached to. The stinger and the moving piece of the shaker to which the stinger is attached have their own mass and stiffness and therefore affect the structure. Shakers are also heavy, and therefore it is often difficult to attach them to the desired place on a structure. Often they are suspended near the structure by elastic cords, which serve to dynamically isolate them as well as to position them. Shakers come in a wide range of sizes. The smallest ones in general use are about the size of a baseball. The largest are called shaker tables, and the intent is to place the entire structure *on* the shaker to seismically excite it. Custom-made shaker assemblies (used in earthquake studies, not surprisingly) can be big enough to shake a small building.

Of course, it's not enough to just excite a structure—we also have to measure the response. The most common device used to measure a structure's response are *accelerometers* (discussed earlier in Section 2.12). These devices are usually piezoelectric in nature and allow us to determine the structure's acceleration as long as we make sure that the natural frequency of the accelerometer is greater than the highest frequency we're interested in measuring. Accelerometers come in a variety of masses, from .5 to 500 g. The mass is clearly of concern if the test structure is light. Since the accelerometer is physically attached to the structure, the mass is increased and must be analytically accounted for if the test results are to be accurate.

It always seems strange to someone new to modal testing, but the most common means of attaching an accelerometer is to simply stick it to the structure with a thin layer of bees wax. Alternatively, one can mount it semipermanently by screwing it to the structure.

A fairly new class of measurement devices is laser based. A laser beam is aimed at a structure, reflected back to the apparatus (called a *laser vibrometer*), and compared with the output beam. The laser vibrometer uses this information to determine what the structure's vibrational motions are like at the target point. Since there is no loading on the structure, this method is an excellent way of examining very lightweight test objects (for which an accelerometer would represent so much additional mass loading that accurate results would be impossible). The major drawbacks to this approach are cost (the instruments start at $15,000–$20,000) and the difficulty of getting the laser signal reflected back to the vibrometer.

An old sensor that is still widely used is the *strain gauge*, a device composed of a material that experiences a resistance change when subjected to strain. By measuring the resistance variation, one can deduce the local strain of the test specimen. The sensors are very cheap but are also relatively noisy and have a limited dynamic range.

Another new way of sensing and actuating a structure is by means of a distributed piezoelectric material. In addition to piezoelectric crystals (used in force transducers and accelerometers), large flat pieces of flexible or rigid piezoactive materials are available. Since the material is distributed over a wide area, one can use it to directly sense particular modes of the structure, or conversely, to excite particular modes. Another variant on this approach is to combine the sensing and actuation into a single piece of piezoelectric material, resulting in a single device that both excites a structure and measures the structure's vibrations.

After a signal has been sensed, it is amplified through an appropriate *signal conditioner* (charge amplifier or voltage amplifier) and then fed to an *analyzer*. The range of analyzers (known as vibration analyzers, modal analyzers, spectrum analyzers, etc.) is very wide nowadays. One can purchase simple two-channel units (one input and one output) as well as sophisticated multichannel ones capable of handling 64 channels or more of data. A few years ago all the analyzers were stand-alone pieces of equipment, but it is becoming more common to see analyzers that work along with a dedicated computer. In this way you can mix and match the amount of analyzer function and computational speed you desire. If you've already got the computer, you can add the analyzer section for a relatively low cost and buy only as many channels of input and output capability as you wish.

All these devices attempt to determine the transfer functions between the input and output signals. For instance, the output might be a sinusoidal signal that is fed to a shaker. The input would be a signal proportional to the structure's acceleration, as measured by an accelerometer. If the test was done at a single frequency, the analyzer would determine the amplitude and phase response of the structure, for this frequency of excitation. If the input was a random signal, the analyzer would use the spectral analysis we've gone over to determine the complete transfer function over a range of frequencies. By analyzing the peak responses, the half-power points, the zeros, etc., the analyzer could also construct estimates of the natural frequencies and damping of the system.

By getting responses from multiple points along the structure, we can build up a database of response information that will allow the analyzer to compute the actual mode shapes of the structure. Analyzers that do this are called *modal analyzers*. Sophisticated units will allow the user to create a model of the system from experimental data and then modify the system to see how changes will affect the overall vibrational response.

8.7 NONLINEAR EFFECTS

A central assumption in all the work so far has been linearity. At the risk of being obvious, this means that all the physical quantities we measure vary in a linear manner. Thus the force in a spring is linearly proportional to the stretch—double the extension and the force doubles. The same is true for the force necessary to accelerate a mass—double the force and mass's acceleration doubles (if there are no additional forces acting on it). Thus a nonlinearity in the system implies that something isn't behaving in a linear manner. One obvious example is a coil spring, like those found

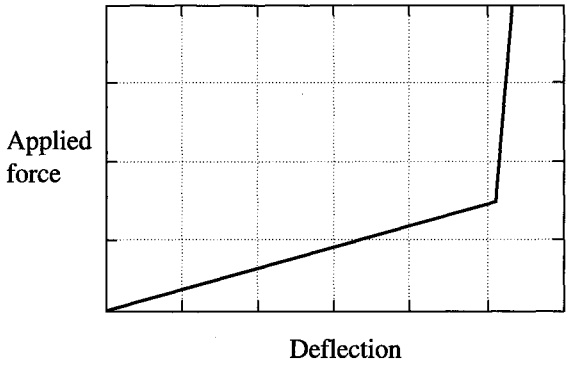


Figure 8.18 Force versus deflection for a coil spring

in automobile suspensions. For small oscillations this spring will behave linearly. What happens, however, if the load in the car becomes so great that the spring is completely compressed and all the coils touch each other? In this case it won't act anything like a linear spring. Additional forces now serve to compress the actual metal of the spring, rather than simply bending a coil. Thus the effective spring constant will be immense. This behavior is shown in Figure 8.18. The deflection versus force shows a normal linear relationship up until the critical deflection, at which point the coils become fully compressed. After this the force versus deflection curve still behaves linearly, but at a much increased spring constant. Physical elements that display such relationships are called piecewise linear. The overall behavior is nonlinear, but in fixed regimes the behavior can be well approximated as linear.

Sources of nonlinearity are common. One example occurs whenever two surfaces rub against each other. The friction between a sliding surface and a fixed surface is given by μN, where μ is the coefficient of friction and N is the normal force. Certainly this isn't linear—it shows no variation at all with velocity, let alone a linearly proportional one. This kind of friction (called Coulomb or dry friction) will always be present in the joints and connections of any structure.

What happens to our linear analyses when an unmodeled nonlinearity is present? If the nonlinearities are small, there's no great effect. However as they become more significant they can substantially alter the modal analyzer's calculations (not unexpectedly). One of the difficulties facing the analyst is trying to determine whether problems with the analyses are due to nonlinearities in the system or to noise. To illustrate how nonlinearities can alter a system's response, look at Figures 8.19–8.23. The system is a simple vibrating beam—with a twist (Figure 8.19), a metal block placed near the beam's midspan. When the block is far from the beam, the beam never touches it, resulting in a linear system. As the restraint is moved closer, it becomes more and more common for the beam to strike the restraint as it is vibrating—a strongly nonlinear effect. Four cases are examined, corresponding to linear, weakly nonlinear, moderately nonlinear, and strongly nonlinear.

Figure 8.20 shows the fully linear response (case 1). Two peaks are present, at approximately 30 and 175 rad/s. A zero is also indicated at about 105 rad/s. This simply means that the output is

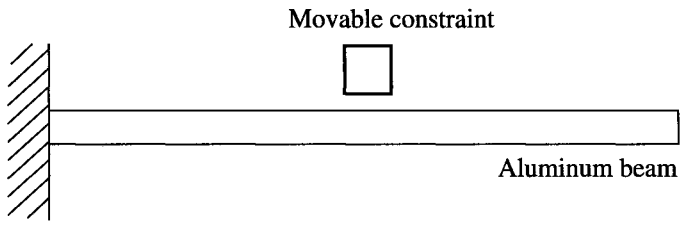


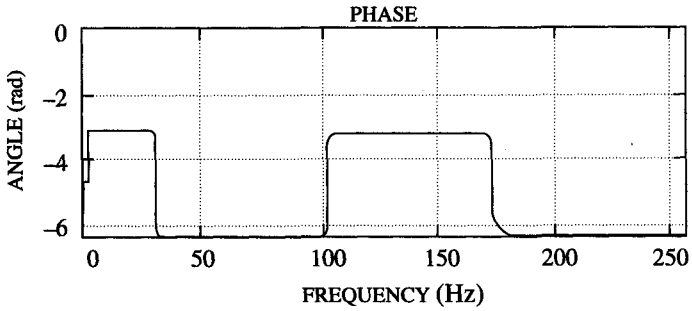
Figure 8.19 Clamped-free beam with rigid constraint

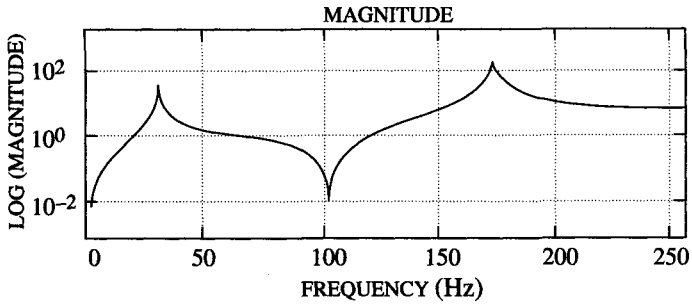
close to zero when the system is excited at that frequency. The phase response is similarly clear. The phase angle versus frequency is very well defined and shows that the system is purely in phase or out of phase, depending upon the frequency. The phase switches around each resonance peak as well as at the zero. Finally, the coherence is excellent. Everywhere away from the two peaks and the zero we see a coherence of 1.0. This result is typical of a good modal test. You shouldn't expect a good coherence around resonances or zeros. At resonance, the amplification effect magnifies any noise in the excitation device, affecting $S_{ff}(\omega)$. When at a zero, the actual response is very small. In this case the measured response will be dominated by sensor noise and $S_{xx}(\omega)$ will be adversely affected. The fact that the coherence is excellent everywhere else is a good assurance that the test results are believable.

In Figure 8.21 (case 2) we show the results of having just a little bit of contact with the block. Because of this contact, we've got a bit of nonlinear contamination. Of course the linear algorithms in the analyzer don't know this, and so they go ahead as if the system were linear. The phase and amplitude responses have changed a bit. The most obvious change is the apparent reduction in damping (shown by the flatter resonance peak) and the "noise" in the phase response. The real indicator that we've got a problem is the coherence results. As you can see, the coherence has degraded quite a bit, especially around the zero (105 rad/s).

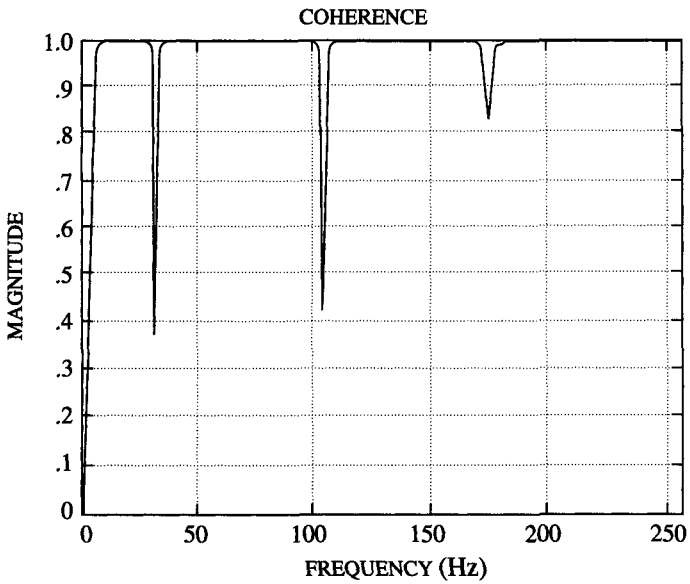
A moderate nonlinearity (Figure 8.22, case 3) causes further degradation. The amplitude response is now very irregular, as is the phase response. Hardly any frequencies show a good coherence response. Finally, Figure 8.23 (case 4) shows the results for a severe nonlinearity. For this case the block is being struck for almost every oscillation of the beam. The results are clearly a mess. The phase response is completely garbled, and we've lost any sense of separate resonance peaks in the amplitude response. As expected, the coherence is terrible, indicating that we shouldn't have any faith at all in these responses.

Table 8.1 shows the first two calculated natural frequencies for the tests in Figure 8.20–8.23, as well as the calculated linear damping approximation. Since the block provides a constraint, and constraints typically stiffen systems, we would expect it to cause the calculated first natural frequency to increase, as indeed happens. Note also the wide variation in the damping estimates. This simply reflects the fact that the measurements are contaminated by the nonlinearity and thus the damping algorithms work badly, trying to find the linear damping response for a system that isn't truly linear.





Magnitude and Phase



Coherence

Figure 8.20 Linear beam response (case 1)

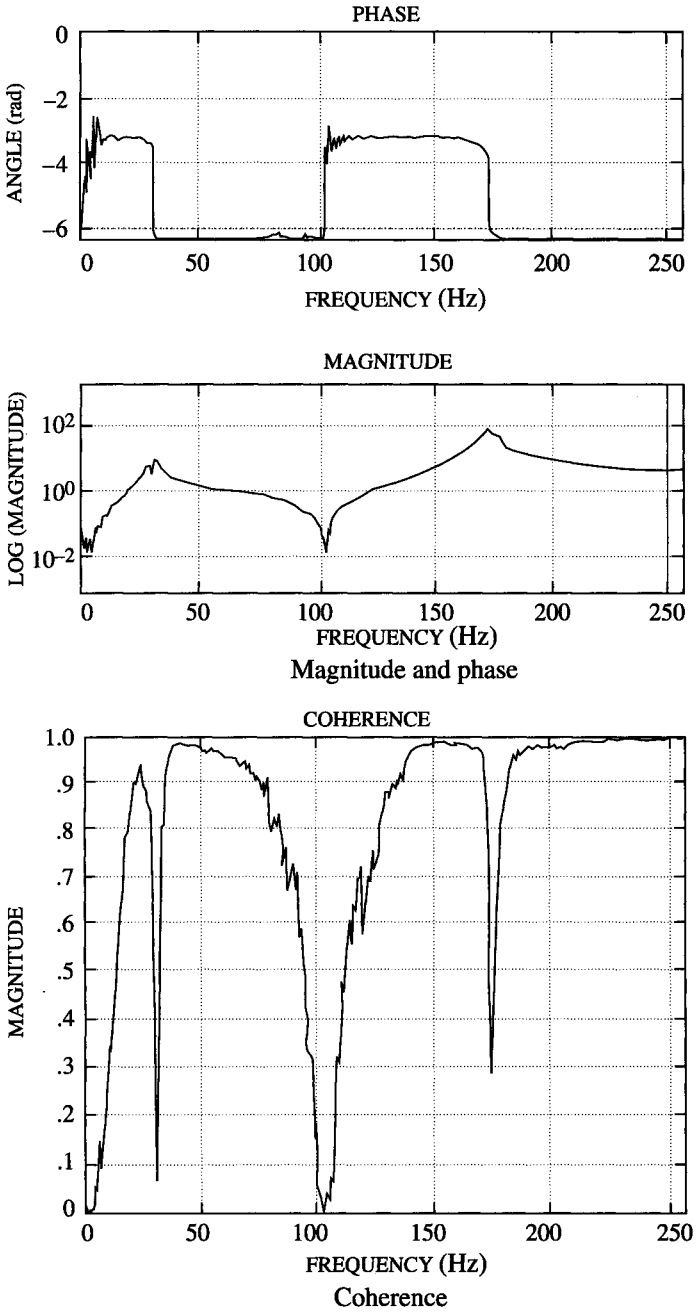
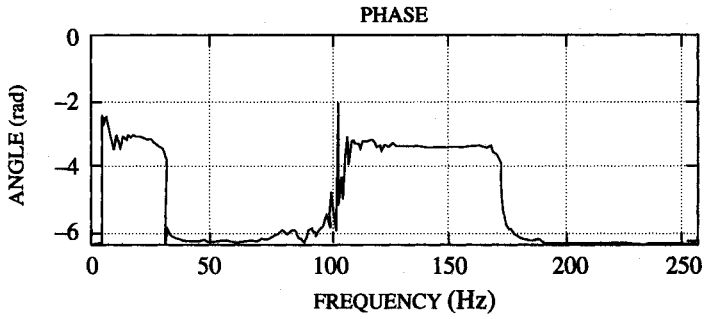
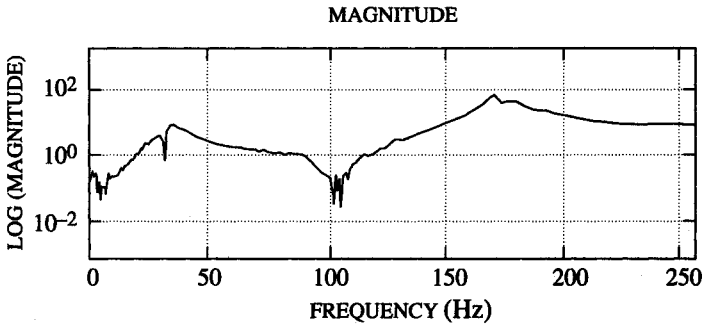
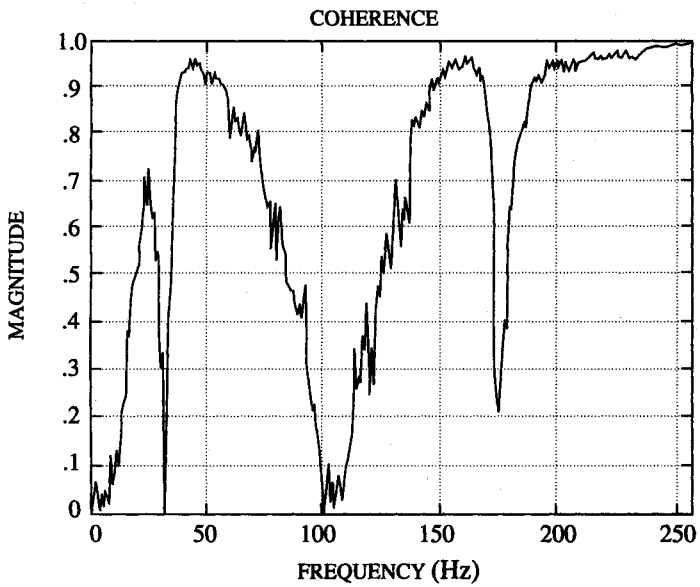


Figure 8.21 Nonlinear beam response (case 2)



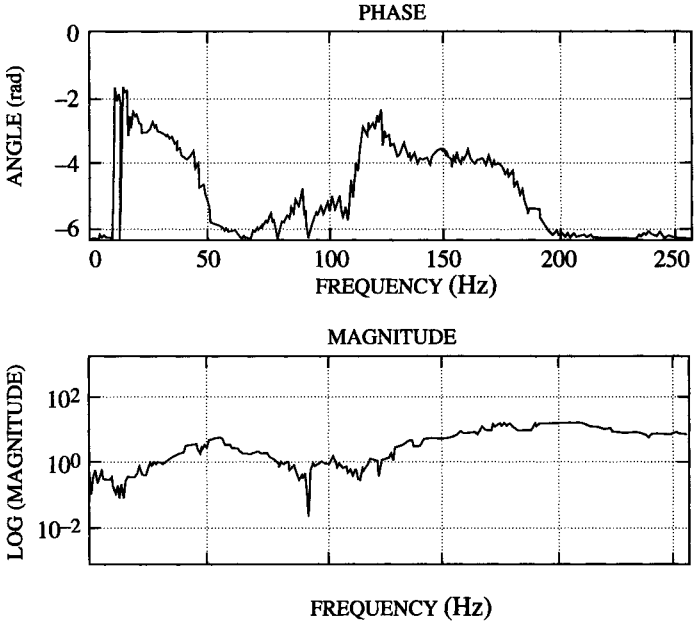


Magnitude and phase

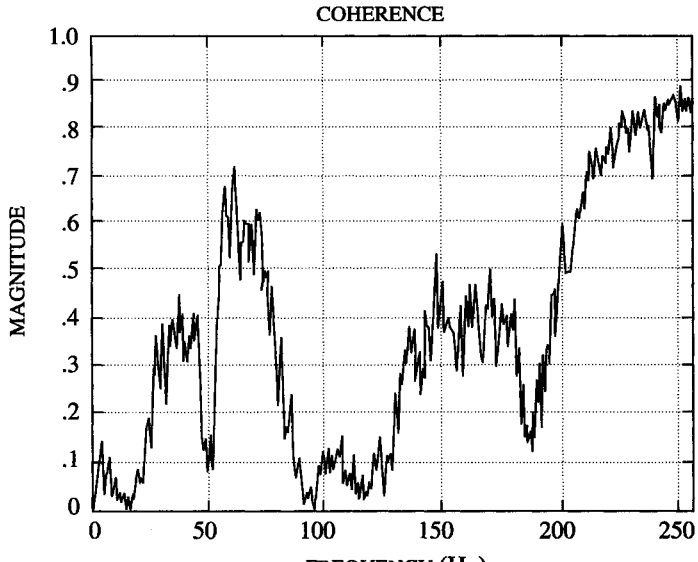


Coherence

Figure 8.22 Nonlinear beam response (case 3)



Magnitude and phase



Coherence

Figure 8.23 Nonlinear beam response (case 4)

TABLE 8.1 FREQUENCY/DAMPING ESTIMATES FOR NONLINEAR BEAM VIBRATION

	Mode 1		Mode 2	
	Frequency	Damping	Frequency	Damping
Case 1	31.2	0.009	174	0.005
Case 2	32.2	0.011	174	0.005
Case 3	34.8	0.003	173	0.020
Case 4	56.0	0.110	174	0.051

8.8 HOMEWORK PROBLEMS

Section 8.2

8.1. Assume that

$$x(t) = \sin^3(t)$$

is the signal of concern. Generate a discrete time series by recording n values of $x(t)$ for

$$t = \frac{2\pi i}{n} \qquad i = 1, \ldots, n$$

Let $n = 10$, 20, and 100. Does the average of these values equal 0 (which is the correct average value for the continuous signal $x(t)$)?

8.2. Repeat Problem 8.1 but let

$$t = \frac{\pi}{4} + \frac{2\pi i}{n} \quad i = 1, \ldots, n$$

That is, include a phase shift. Does the average come out to zero? Why?

8.3. Repeat Problem 8.1 but take the average of a 20-second interval, i.e., average

$$x(t) = \sin^3(t)$$

over $0 \leq t \leq 20$. Use a 20-point discretization. Does the average equal zero? Why? How does this impact a real-world data analysis?

8.4. Repeat Problem 8.1 but vary the averaging period. Take the average over 10, 20, and 100 seconds. To keep the resolution constant, use 10 data points for the 10-second average, 20 for the 20-second average, and 100 for the 100-second average. Does the average for each trial equal zero? How do longer averaging times affect the average? Comment on the implications for data analysis.

8.5. Can you find a finite averaging time for which a discrete sampling of

$$x(t) = \cos(t) + \sin(2t)$$

will average to zero? What about

$$x(t) = \cos(t) + \sin(\sqrt{3}t)$$

8.6. Calculate the variance of

$$x(t) = 2\sin(4\pi t) - 5\cos(2\pi t)$$

over the time interval $0 \leq t \leq 1$. Demonstrate numerically that the variance of the sum of the two signals is equal to the sum of the variance of the individual components.

8.7. Problem 8.6 showed that the sum of the variance is equal to the variance of the sum when several signals are being added together. Examine the same function:

$$x(t) = 2\sin(4\pi t) - 5\cos(2\pi t)$$

and compute the variance for 20, 200, and 600 data points. Use a sampling interval of .0415 second. Explain the results.

Section 8.3

8.8. Compute the complex Fourier transform of the $x(t)$ shown in Figure P8.8.

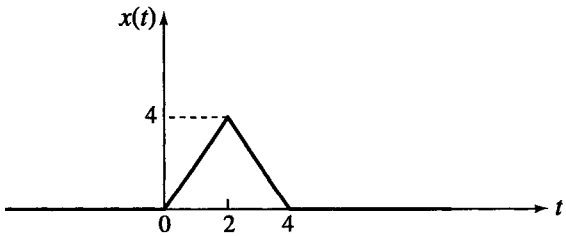


Figure P8.8

8.9. Compute the complex Fourier transform of the $y(t)$ shown in Figure P8.9. How does the answer differ from that of Problem 8.8? Explain how you can obtain the answer to Problem 8.9 if you already know the answer to Problem 8.8 (i.e., without explicitly calculating the Fourier transform).

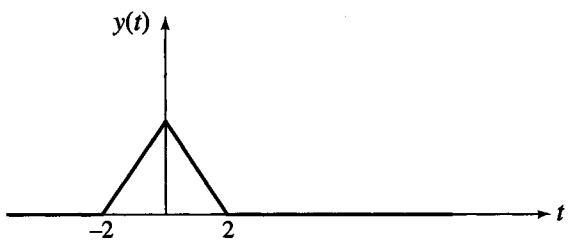


Figure P8.9

8.10. Show that the Fourier transform of a signal $y(t)$, where $y(t)$ is simply a delayed version of a signal $x(t)$, is equal to the Fourier transform of $x(t)$ multiplied by $e^{-i\tau\omega}$, where τ is the time delay.

8.11. Show that the complex Fourier transform is purely imaginary if the time function is odd and the transform is purely real for even functions. (*Hint*: Consider what the Fourier series representation of odd and even functions looks like.)

8.12. Calculate the Fourier transform of $x(t)$ and $y(t)$ shown in Figure P8.12. How does the delay in time appear in the Fourier transform?

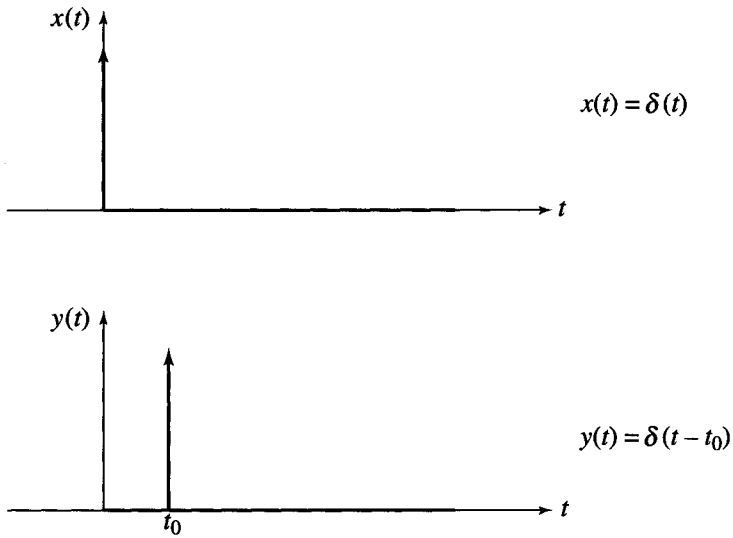


Figure P8.12

Section 8.4

8.13. Find the solution to $\ddot{x} + .2\dot{x} + 4x = \delta(t-1)$; $x(0) = 0$, $\dot{x}(0) = 0$.

8.14. Find the solution to $\ddot{x} + .2\dot{x} + 4x = \delta(t-1)$; $x(0) = 1$ cm, $\dot{x}(0) = 0$.

8.15. Find the solution to $\ddot{x} + .2\dot{x} + 4x = \delta(t)$; $x(0) = 0$, $\dot{x}(0) = 2$ m/s.

8.16. What does the impulse response look like for an overdamped system?

8.17. Calculate and plot $R_{xx}(\tau)$ for the power spectral density illustrated in Figure P8.17. Let $\omega_1 = 1.6$ rad/s and $\omega_2 = 2$ rad/s.

8.18. How does the answer to Problem 8.17 change as ω_1 goes toward ω_2? (Keep the total energy constant.)

8.19. Consider the system shown in Figure P8.19. Let the forcing be described by band-limited white noise. Such an excitation has a power spectral density like that shown in the Figure P8.19. The input is unpredictable and has energy from $\omega = 10$ to 100 rad/s. Determine the mean-square value of the system's response. $m = 1$ kg and $k = 90$ N/m.

8.20. Verify your answer to Problem 8.19. Break the forcing spectra into segments of width $\Delta\omega$ as shown in Figure P8.20. Using the information in the text, determine what a delta function of magnitude

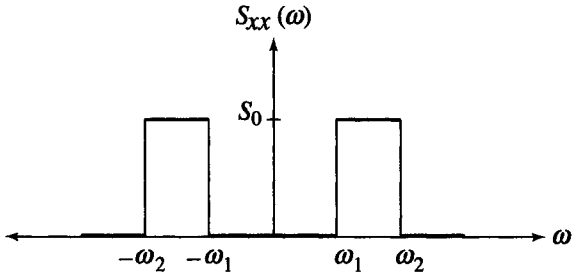


Figure P8.17

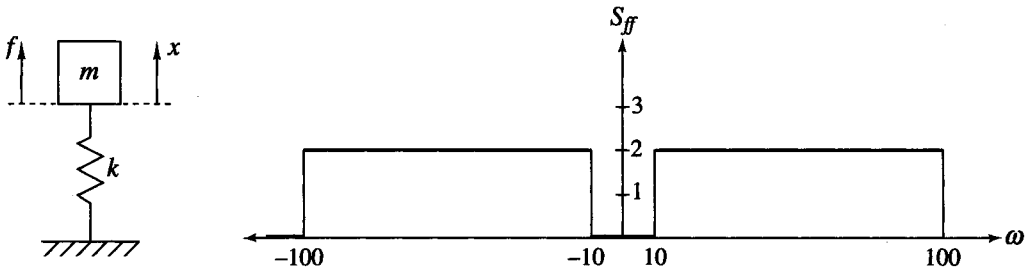


Figure P8.19

equal to $2\Delta\omega$ and centered at $10 + \frac{\Delta\omega}{2} + i\Delta\omega$ (i.e., the center points of s_1, s_2, s_3, etc.) corresponds to in the time domain (*Hint*: It corresponds to a sine wave of frequency $10 + \frac{\Delta\omega}{2} + i\Delta\omega$ with a magnitude that remains for you to determine.) Calculate the mean-square response of the system to each sine and sum the responses. This should give you the same result you found from Problem 8.19, since we're just approaching the problem from the time domain rather than the frequency domain.

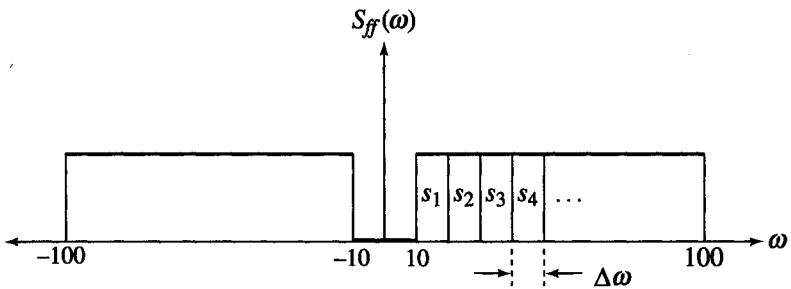


Figure P8.20

8.21. In this problem we'll show an alternative way to numerically evaluate the answer to Problem 8.19. As in Problem 8.20, break S_{ff} into segments and approximate the input by a string of δ functions (Figure P8.21):

$$\omega_1 = 10 + \frac{\Delta\omega}{2}$$
$$\omega_2 = 10 + \frac{3\Delta\omega}{2}$$
$$\omega_3 = 10 + \frac{5\Delta\omega}{2}$$
$$\vdots$$

Evaluate the spectral integral relating $S_{xx}(\omega)$, $\frac{1}{k-m\omega^2}$, and $S_{ff}(\omega)$ by utilizing the δ approximation and show that you obtain the same answer as in Problems 8.19 and 8.20.

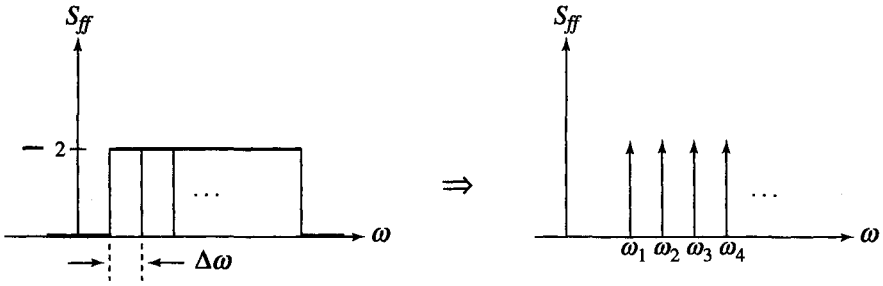


Figure P8.21

8.22. Determine the autocorrelation function for the $x(t)$ shown in Figure P8.22.

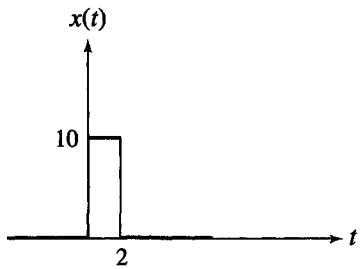


Figure P8.22

8.23. How does the autocorrelation change from that of Problem 8.22 if the rectangular function is repeated as shown in Figure P8.23?

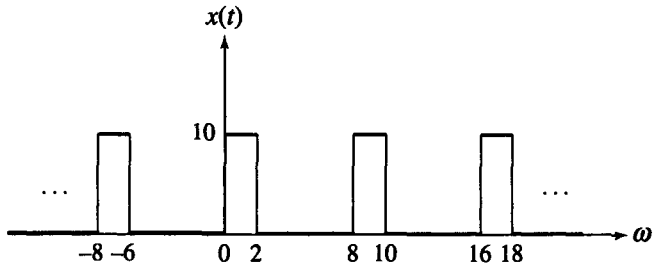


Figure P8.23

8.24. Determine the autocorrelation function for the function having the power spectral density function shown in Figure P8.24. The magnitude of each delta function is 20. What kind of signal does this correspond to?

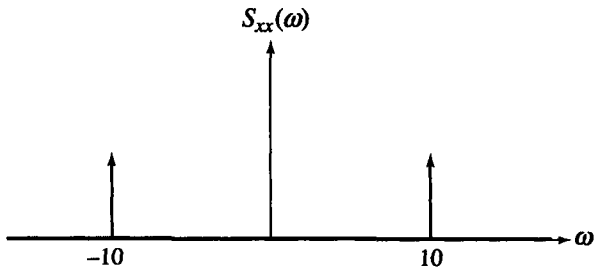


Figure P8.24

8.25. Use Figure P8.25 to determine the autocorrelation function for the function having the following power spectral density function:

$$S_{xx}(\omega) = 20\delta(\omega - 10) + 20\delta(\omega + 10) + 40\delta(\omega)$$

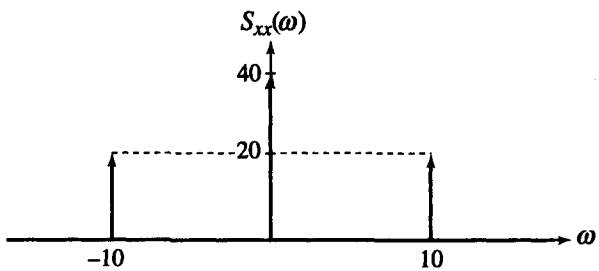


Figure P8.25

8.26. Find the mean-square response of a system having a transfer function

$$g(\omega) = \frac{2}{8 + 2i\omega}$$

for an ideal white noise input; i.e., the input has a power spectral density equal to

$$S_{ff}(\omega) = S_0$$

8.27. How does the answer to Problem 8.26 change if the input is band limited so that

$$S_{ff}(\omega) = S_0; \quad -7 \leq \omega \leq 7$$
$$S_{ff}(\omega) = 0; \quad |\,\omega\,| > 7$$

Section 8.7

8.28. If you're concerned about nonlinearities in your system, you might use an impact hammer and tap the structure lightly, identify the system parameters by means of a modal analyzer, and then repeat the procedure with a more forceful hammer strike. How could you use the resulting information to determine whether the structure is acting nonlinearly?

8.29. Numerically integrate

$$\ddot{x} + x + x^3 = 0; \; \dot{x}(0) = 0$$

and compare the frequency of the resultant oscillations for $x(0) = .1$, $x(0) = 0.2$ and $x(0) = 0.6$. Does the nonlinearity act to "stiffen" the system (raise the natural frequency or "soften" it (lower the natural frequency)? Plot the force versus deflection curve for the equivalent spring

$$F = x + \varepsilon x^3$$

and try to explain why the system behaves as it does.